Maurizio Ungaro

Research and Software Resume

Staff Scientist, Jefferson Lab | Nuclear physicist and simulation software developer
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Summary

Staff Scientist at Jefferson Lab with experience spanning nuclear physics, detector systems, Geant4 simulation, GEMC/GEMC3 and pygemc development, validated CAD geometry, and distributed CLAS12 production workflows.

Core Strengths

- Build and support Geant4-based simulation workflows for detector and radiation-transport studies.
- Develop GEMC3 and pygemc; migrate CLAS12 geometry and detector plugins to the new framework.
- Repair and optimize CAD meshes with construction, overlap, and tracking benchmarks.
- Support CLAS12 detector operations, including LTCC calibration, maintenance, and performance studies.
- Translate complex scientific software into tutorials, examples, technical notes, and user-facing documentation.
- Teach and mentor students in laboratory instruction, data analysis, detector calibration, and scientific computing.

Selected Experience

Staff Scientist 2011–present
Jefferson Lab, Hall-B
GEMC/GEMC3 and pygemc development; CAD validation; simGrid/OSG production support; LTCC detector work; nucleon-structure analysis.

Research Associate 2004–2011
University of Connecticut
CLAS analyses, meson electro-production, CLAS12 software, and detector simulation development.

Selected Projects

GEMC / GEMC3	Geant4 framework development, Qt visualization, localization, and XR integration.
pygemc	Python geometry tools, PyVista rendering, and analysis workflows.
Geant4 at JLab	Tutorials, examples, and support material for Jefferson Lab users.
CLAS12 Simulations	Tagged releases, GEMC3 migration, CAD repair, and validation benchmarks.
CLAS12 on OSG	simGrid production workflows and queue management using HTCondor and OSG.
Low Threshold Cherenkov Counter	Forward detector operation, calibration, and performance for pion/kaon discrimination.
Heavy Photon Search	Dark-sector search collaboration at Jefferson Lab.
SPring-8	Prior experimental work in Japan through LEPS/SPring-8 collaboration.

Technical Skills

Simulation: Geant4, GEMC3, pygemc, CLAS12, ROOT, CAD validation, PyVista, Qt.
Programming: C++20, Python, shell, Git, CI, Docker, Meson, CMake, SCons.
Computing: HTCondor, OSG, SQLite, CSV/JSON/ROOT workflows, reproducible environments.
Communication: documentation, tutorials, talks, mentoring.

Education

Ph.D. in Nuclear Physics, Rensselaer Polytechnic Institute.
Laurea in Fisica, Universita degli Studi di Genova.

Teaching and Mentoring

Adjunct Assistant Professor of Physics, Christopher Newport University, 2011. Instructor for PHYS 105 and PHYS 152 laboratories. Teaching assistant at Rensselaer Polytechnic Institute for Physics I and Electromagnetic Theory. Mentor for undergraduate and graduate research in Jefferson Lab data analysis, detector calibration, and CLAS/CLAS12 physics.

Profiles

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